

Lab 1 Task 2 - Scientific Writing

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1 Introduction

This lab follows the scientific writing course at Duke University. The exercises are also extracted from the course website.

You should revise the following sentences for clarity using the information explained on the following website:

- <https://cgi.duke.edu/web/sciwriting/index.php>

For each sentence, you should justify your answer.

2 Subjects and Actions

1. We subjected yeast to 20 min of amino acid deprivation and made ribosome-footprint and mRNA-abundance measurements.

We deprived the yeast of amino acids for 20 minutes and then measured ribosome footprint and mRNA abundance.

Sequential events, stronger verb usage, removed unnecessary words.

2. The number of different mechanisms that may exist for cells to interpret morphogens, and the importance of design features such as feedback or local cell-cell communication, is unclear.

How many mechanisms cells use to interpret morphogens and how feedback and local cell-to-cell communication contribute to this process remain unclear.

The original sentence doesn't convey what it wants exactly and is easily misunderstood. Removed unnecessary words. Emphasize the purpose of the sentence to say that it's still unclear why these things are happening.

3. Localized fluctuations in substitution rate are widely employed to draw inferences concerning the phenotypic significance of genomic sequence.

Researchers widely use localized variation in substitution rates to infer the phenotypic significance of genomic sequences

More concise, better readability, and uses active voice.

4. In this study, we subjected seven different primate species to comparative analysis of the radial distribution pattern of human chromosome 18- and 19-homologous chromatin by three-dimensional fluorescence in situ hybridization.

Researchers used three dimensional fluorescence in situ hybridization to compare the radial distribution patterns of chromatin homologous to human chromosomes 18 and 19 across seven primate species.

Removing redundant words, present tense wording. Centering the approach rather than the testing subjects.

3 Cohesion, Coherence, and Emphasis

1. These data were then subjected to hierarchical clustering using the Pearson correlation as the distance metric.

We then hierarchically clustered these data using Pearson correlation as the distance metric

Passive voice. New information last.

2. We designated the putative GRB target genes as the human TFs that were under a HCNE density peak and whose orthologs were in conserved synteny with an array of HCNEs in human:zebrafish alignments. Density peaks and syntenic regions were downloaded from the Ancora Genome Browser.

HCNE density peaks and syntenic regions was downloaded from the Ancora Genome Browser. Human transcription factors as putative GRB target genes was designated to see if they occurred under an HCNE density peak and their orthologs retained conserved synteny with an array of HCNEs in human:zebrafish alignments.

Order reversion via passive voice. new information last. Matching first and last sentences in the paragraph.

3. Improvements are expected in the predictive power of all the scores being computed on multispecies alignments.

We expect multispecies alignments to improve the predictive power of all the computed scores.

Clear invocation that improvements are expected. The outcome improves the predictive power gets greater emphasis.

4. With the development of molecular embryology and the coming of the post-genomic era, the molecular mechanisms of morphological evolution have recently begun to be elucidated.

Advances in molecular embryology and post-genomic research have recently begun to elucidate the molecular mechanisms underlying morphological evolution

Passive voice as we don't know who advanced the method. Starts with scientific development and ends the sentence with new information.

4 Concision and Simplicity

1. One of the complexes formed by the hematopoietic transcription factor Gata1 is a complex with the Ldb1 and Tal1 proteins.

Gata1 forms complexes with the proteins Ldb1 and Tal1.

Removes repetition and complex wording. As well as using simpler words.

2. A phylogenetic tree based on the number of differentially expressed genes between species recapitulates their known phylogeny.

A tree based on the differentially expressed genes between species reproduces their known phylogeny.

Remove the repetition of phylogeny, needlessly complex word for reproduces: "recapitulates".

3. It is important to note for some of our later discussion that we permit a block to have only one row.

A block may only contain one row

Removes the empty introduction in the sentence.

4. These sequences are generally among the least constrained in the genome, so they constitute a plausible neutral standard accounting for regional variation in mutation and recombination rates.

These sequences are usually among the least constrained in the genome. They therefore provide a plausible positive neutral standard for regional variation in mutation and recombination rates.

Divides a longer sentence into smaller ones. Uses simpler words. Still keeps adjectives that convey necessary distinctions.